

June 8, 2026

Editors & Reviewers
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Re: Manuscript Revision 1

Dear Editors and Reviewers,

We wish to thank you for the careful and constructive review of our manuscript titled "From Where to What: The Geo-intervention Modelling Framework" (Manuscript ID: *ijgi-4299239*). The feedback was very helpful and has led to valuable improvements to our manuscript.

Please find a point-by-point response to each comment submitted on the review system (copied below), with all changes visible in red in the accompanying tracked changes word document.

Sincerely,

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On behalf of all co-authors

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Reviewer 1

Comment 1.1: This paper deals with a very current topic in the field of GIScience and spatial decision support systems, namely how spatial data and predictive models can be used not only to identify spatial problems, but also to generate concrete interventions. I believe that the main idea of the article is interesting and well chosen, especially by trying to answer the question "what interventions should be applied", not just "where do the problems occur".

The paper is well organized, and the authors gradually build the conceptual framework starting from the relationship between actions, locations, and outcomes. I found it useful that this relationship is also formulated mathematically, which provides methodological coherence to the entire framework. Also, the Toronto case study is relevant and demonstrates that the framework can be applied to large and heterogeneous geospatial data sets.

Another positive aspect is the integration of AutoML and spatial optimization components into a unified workflow. In particular, the use of multiple urban infrastructure categories and their aggregation into regular grids is a practical and easy-to-follow approach.

However, I believe there are a few issues that should be clarified before publication.

Response 1.1: We are grateful for the kind and encouraging assessment, thank you. It is especially reassuring to hear that the mathematical formulation added methodological coherence and that the integration of AutoML with spatial optimization components came across as a practical and accessible approach. These were deliberate design choices intended to support both technical and interdisciplinary readers, so the positive reception is again, reassuring. The specific concerns raised in the subsequent comments have each been carefully addressed, with all changes visible in red in the attached tracked changes document (automatically named *author-coverletter-55869788.v1.docx*).

Comment 1.2: First, the validation part remains quite limited. The authors correctly mention the difficulty of verifying geo-interventions in real conditions, but in its current form the results remain predictive rather than empirically proven. I think the paper would be stronger if there was a more detailed discussion of the difference between correlation and causation, especially in the context of AutoML models.

Response 1.2: We thank the reviewer for this important observation. We agree that a more detailed treatment of the distinction between predictive association and causal effect would strengthen the paper, and have added a dedicated subsection (Section 4.3.3, Causal Inference, under the discussion) to address this directly. The subsection acknowledges that the outcome modelling component learns associations from observational data rather than estimating causal effects, and that variable importance reflects predictive contribution rather than interventional effect. In the case study, the scenario two is discussed as an illustration of the model's unreliability when applied beyond its training distribution. The subsection also suggests utilizing the framework's outputs as hypothesis candidates requiring causal validation, and outlines both the quasi-experimental designs and causal machine learning approaches that could ground future iterations of the framework in estimated causal effects.

Comment 1.3: Also, some variables identified as important raise some interpretative questions. For example, the relationship between transit shelters, wayfinding structures and the reduction of road collisions is not sufficiently explained. It is currently not very clear whether these elements contribute directly to the reduction of accidents or whether they indirectly reflect urban characteristics associated with other types of infrastructure and traffic.

Response 1.3: Thank you for raising these concerns. We appreciate the thoughtful insights on interpreting the case study results. Section 4.3.3 also addresses this directly. Transit shelters and wayfinding structures are discussed as variables that are associated with MVCs in the training data, with the explanation that these elements likely serve as proxies for high-activity urban areas where collision exposure is inherently greater, rather than directly influencing collision risk. This indirect, exposure-mediated relationship is precisely the kind of confounding that makes predictive variable importance an unreliable guide to intervention, and the revised section uses these variables as illustrative examples of the broader causal limitation of the framework.

Comment 1.4: The part on model performance could also be developed. MAE values for different grid resolutions are presented, but details on model stability, possible overfitting or comparison with simpler methods are missing. An additional discussion on model robustness would increase the credibility of the results.

Response 1.4: We appreciate this observation and agree that details on model stability, overfitting, and model robustness would strengthen the credibility of the results. We have added two sentences to Section 2.3.2 (Outcome Modelling Approach) in the tracked changes document noting that both *auto-sklearn* and *tpot* employ internal cross-validation during their respective automated searches to mitigate overfitting. We also draw attention to Appendix B.2, which addresses the broader question of baseline and hybrid model selection, and explicitly recognizes that a hybrid approach combining trusted prior-knowledge baseline models with more automated approaches can produce stronger empirical evidence for model selection. The choice to apply two independent AutoML packages with different internal search strategies additionally provides a degree of cross-method validation, as both packages were evaluated, and the better-performing result was selected. As the case study was simplified to demonstrate application and provide understanding of the geo-intervention framework, we did not wish to draw attention away from it and complicate the focus of the paper.

Comment 1.5: The discussion section is well written and I found it to be one of the strongest parts of the article, especially in how it addresses the limitations of the framework and the role of human expertise in decision-making. However, some passages about LLMs and future developments are quite speculative and could be reduced to keep the focus on the main contribution of the paper.

Response 1.5: Thank you for the encouraging comment. We agree that LLMs and future developments are speculative. However, they cover very recent developments and illustrate a meaningful future research direction for the framework. We have revised Section 4.2.3 (Contextual Reasoning) in the

discussion to explicitly characterize LLM integration as an open and speculative area of research, noting that LLMs do not currently reason or operate at the human level for all tasks. Rather than reducing this content, we have also expanded the section to situate LLMs within the broader landscape of contextual reasoning approaches. Specifically, we now note that more established and empirically demonstrated approaches, including knowledge-based systems integrated with GIS, Explainable AI methods such as SHAP and LIME, and ontologies and knowledge graphs, represent more immediately actionable avenues for incorporating contextual reasoning into the framework in its current form. We have additionally revised Section 4.4.1 to reflect that LLM integration may improve contextual reasoning but remains an open research direction that warrants empirical investigation. We hope that this now conveys a stronger emphasis on the contextual reasoning of the framework, rather than solely LLMs.

Comment 1.6: Overall, I believe the article is valuable and proposes an interesting direction for research. The framework presented has potential for future applications in GIS and urban analytics, but the paper would benefit from further clarification regarding the validation and interpretation of the results.

Response 1.6: We are glad the overall direction of the research was received positively. The comments made around validation and result interpretation were very helpful for our manuscript, and we hope the revisions have addressed all your concerns. Thank you again for your time and efforts to review our manuscript.

Reviewer 2

Comment 2.1: In this study, a new framework called the “Geo-intervention Modelling Framework” is proposed to identify not only where problems occur in cities, but also what kinds of interventions may reduce them. The authors combine spatial data, machine learning, and optimization methods to analyze traffic collisions in Toronto, Canada. The topic is important for GIScience and spatial decision support systems. The manuscript is generally well-structured, but several issues should be revised to improve its technical quality and clarity. My comments are listed below.

Response 2.1: We appreciate the constructive and thorough review of the manuscript and are happy to know that the study's overall structure was clear. Each of the specific technical and clarity concerns raised has been carefully considered and addressed in the responses below, with all changes visible in red in the attached tracked changes document (automatically named *author-coverletter-55952200.v1.docx*). Thank you for your time and effort in reviewing our manuscript.

Comment 2.2: The novelty of the study is not clearly explained. The manuscript claims that existing GIS studies primarily focus on where interventions should occur rather than on which interventions should be applied. However, this claim is not sufficiently supported. The study currently appears to combine existing methods within a single framework. Therefore, the authors should explain more clearly how the study differs from the existing literature.

The authors should also clarify the study's main novel contribution. It should be specified whether the novelty is conceptual, methodological, or integrative.

Response 2.2: Thank you for noting this. We appreciate you bringing this to our attention as it would drastically improve the readability and understanding of our introduction section. We have added four sentences to the last paragraph of the Introduction section in the tracked changes document. The first sentence re-iterates the gaps in literature covered by three comprehensive systematic reviews of the field already cited in the manuscript, clarifying the basis for the novelty claim explicitly. The rest of the sentences further clarifies the nature of the novelty in terms of conceptual and integrative contributions. We hope that these changes reflected in the tracked changes document clarify the novelty of the study clearly.

Comment 2.3: The study treats important model variables as direct recommendations for intervention. For example, red-light cameras are optimized because they are important predictors in the model. However, predictive importance does not necessarily indicate causality. Therefore, the distinction between prediction and causation, as well as the limitations of feature importance.

Response 2.3: Thank you for identifying this as it highlights an important limitation of our framework. We have provided an additional subsection under the discussion (Section 4.3.3, Causal Inference) that addresses this limitation. The subsection explicitly states that treating permutation importance from a predictive model as a direct guide to effective intervention is subject to an important flaw, as the variables with the highest predictive importance may be associated with the outcome through confounding rather than a direct causal pathway. Red light cameras are used as a specific illustrative example: their association with MVCs in the training data most likely reflects placement policy and biases, whereby cameras are installed disproportionately at already high-collision intersections. The subsection makes clear that the framework's optimization step identifies patterns in historical co-occurrence rather than estimating the effect of physically changing the infrastructure. This reframes the generated geo-interventions as hypothesis candidates rather than causal recommendations.

Comment 2.4: The optimization framework appears theoretically well-defined, but in practice, it does not appear sufficient. Provide sufficient information related to this issue.

Response 2.4: We appreciate this concern and acknowledge that the optimization framework, as presented, is intentionally introduced at a theoretical and demonstration level, as is common practice when introducing methodological frameworks before empirical evidence accumulates across applications. We draw attention to Section 4.3.2 in the tracked changes document, which explicitly discusses the limitations of the framework in terms of real-world verification, noting that the lack of available intervention data made it impractical to verify the feasibility and effectiveness of the generated geo-interventions at the time of the study. We further direct attention to an additional Section 4.3.3, which addresses the important distinction between predictive associations and causal effects, and to Section 4.1.3, which discusses the role of practitioner and domain knowledge in guiding the framework towards more feasible and practical geo-interventions. In terms of practical efforts within the optimization itself,

we have added text to Section 2.2.3 and Section 2.3.3 clarifying that action value bounds were derived dynamically from the observed data range of each variable, and have added a note to Appendix B.3 acknowledging physical capacity constraints as an important practical consideration for guiding the optimization algorithm.

Comment 2.5: In the study, several concepts were repeated across the study. The term of actions was repeated 75 times, locations were repeated 65 times, location was repeated 25 times, and outcomes were repeated 80 times. This repetition may reduce readability and weaken the manuscript's impact. Thus, they should be revised, and the number of these terms (if possible) should be reduced.

Response 2.5: Thank you for your time and effort to bring this observation to our attention. We have considered this carefully and decided to maintain the current usage of the terms to encourage accessibility amongst interdisciplinary readers of the article. Actions, locations, and outcomes are the three foundational concepts of the geo-intervention modelling framework, introduced with definitions in Section 2.1 and mapped to mathematical notation throughout ($A_{i,j}$, L_i , and Y_i , respectively). The paper employs several variables and consistently restating the plain-language term alongside or in place of notation is an intentional design choice to support readability for the interdisciplinary audience of IJGI, which includes GIS practitioners and domain experts who may not work routinely with mathematical formalism. Replacing the defined terms with notation alone would require readers to hold the full variable mapping in working memory across our manuscript, which we consider a greater cognitive burden than encountering a defined term repeatedly. However, we genuinely appreciate the time and effort you have taken to bring this our attention, and we wish to acknowledge that there remains repetition of these terms that may not be to the preferences of some readers, potentially hindering the reading experience for these readers, and we apologize for this inconvenience in advance.

Comment 2.6: The authors acknowledge that validation could not be performed because of insufficient data. However, this is one of the study's major weaknesses. The manuscript demonstrates that a predictive model and optimization framework can be constructed, but it does not show whether these interventions would actually reduce traffic collisions in real-world conditions.

Response 2.6: Thank you for identifying this as one of the study's major weaknesses. We agree that this is a critical gap that must be directly addressed. We have added Section 4.3.3 (Causal Inference) to directly confront the distinction between predictive associations and causal effects, acknowledging that the outcome modelling component learns associations from observational data rather than causal effects, and that the generated geo-interventions therefore represent hypothesis candidates requiring causal validation rather than prescriptions for action. In addition, we have expanded Section 4.3.1 (Geo-intervention Verification) to outline several approaches through which such real-world validation of the framework's outputs could be pursued. These include: (1) retrospective back-testing using a temporal holdout; (2) treating independently occurring, policy-driven infrastructure deployments as natural experiments, using a spatial difference-in-differences design; and (3) prospective approaches ranging from small-scale pilot implementations to systematic phased rollouts across spatial clusters. Together, we

hope these additions directly acknowledge the validation gap and provide a clear research agenda for establishing real-world effectiveness.

Comment 2.7: Different grid systems were used, and model performance improved as the number of grid cells increased. However, this issue is discussed only briefly and should be examined in more detail.

Response 2.7: Thank you for this observation. We agree that this requires more detail. In response, we have expanded Appendix B.1 (Spatial Feature Engineering Considerations) of the tracked changes document to include a discussion of the Modifiable Areal Unit Problem (MAUP) as a consideration when selecting spatial aggregation units, explaining how different spatial unit sizes and boundary configurations can produce different results and how this applies to the framework, influencing variable relationships learned and geo-interventions produced. The Toronto case study was used as an illustrative example, noting the variation in model performance across the three grid resolutions evaluated, and identifying multi-resolution sensitivity analysis as an important consideration when applying the framework.

Comment 2.8: Some figures contain many components and conceptual flows. Although informative, they are visually dense and difficult to follow. Simplifying these figures may improve readability.

Response 2.8: Thank you for bringing this concern to our attention. We have carefully reviewed all figures in the manuscript. Although we do agree that simpler figures would improve readability, introducing an additional figure on top of the two figures may distract readers, while removing any of these figures causes a loss in detail that may reduce clarity in understanding the framework. Figure 3 presents a simplified high-level overview of the three framework components and their sub-components, designed to be accessible to a broad interdisciplinary readership, while providing clarity and reference of all components defined in the methodology. Figure C1, provided in Appendix C, is intentionally more detailed as it illustrates the application of the framework to the Toronto case study, requiring additional specificity to convey how each framework component maps to the case study inputs, models, and outputs. We consider each figure to serve a distinct and necessary purpose, and further simplification may risk removing content essential to either conceptual understanding or case study transparency. However, we appreciate the observation and again, thank you for taking the time and effort to bring this to our attention.

Comment 2.9: There are some statements, such as “most studies” that should be supported with appropriate references.

Response 2.9: Thank you for the thorough reading. Upon careful review, we identified two statements without appropriate supporting references or required rewording for clarity. In Section 4.1.1, we reworded the sentence beginning "As most models do not hold knowledge or reasoning of feasibility..." to be "As predictive models commonly do not inherently encode knowledge or reasoning about feasibility in relation to decision-making, knowledge from practitioners and experts is invaluable in adjusting the geo-interventions for feasibility, as exemplified in GIS-based spatial modelling contexts

[109,110].”. We scoped this to predictive models, reframed it as a common limitation, and specifically noted it was exemplified in GIS-based spatial modelling contexts. In Section 1 (the introduction), we included additional references in sentence “For example, spatial clustering techniques are used to locate road traffic collision hotspots [27], but what actions can be taken to decrease road traffic collisions at these hotspots (e.g., deploying traffic enforcement devices [28], reducing road widths [29–32]) are often left to decision makers and experts to identify prior or post analysis [11,26]”. Particularly, references were added for spatial clustering techniques for road traffic collision hotspots, deploying traffic enforcement devices, and road width reductions, while references were included for the statement related to decision makers and experts commonly identifying actions to take prior or post analysis.

Reviewer 3

Comment 3.1: The manuscript proposes a computational framework for geo-interventions. It integrates spatial data processing, automated machine learning, and Bayesian optimization to generate optimized infrastructure changes, so as to improve urban outcomes. The topic is very interesting and is consistent with the scope of the journal.

I would recommend the authors address the following concerns before next stage.

Response 3.1: We appreciate the interest and are glad the manuscript was found to be a good fit for the journal's scope. Each of the concerns raised has been carefully considered and addressed in the responses below, with all changes visible in red in the attached tracked changes document (automatically named *author-coverletter-55952778.v1.docx*). Thank you for your time and effort in reviewing our manuscript and providing feedback with detailed references.

Comment 3.2: The Introduction (which involves literature review) section should be more organized. The first paragraph (Lines 29-69) is excessively long. The distinction between geo-intervention and existing terms like Spatial Decision Support Systems should be more concise and more highlighted.

The authors actually involved a very rich collection of references, which is good. If necessary, a separate Literature Review section can be introduced.

Response 3.2: Thank you bringing this to our attention. We have split the opening paragraph to create three more focused paragraphs: the first establishing the motivation and computational opportunity for geo-interventions, the second summarizing the gap in existing approaches, and the third providing further evidence of the gaps as noted by extensive literature reviews. We have also added a sentence before the last sentence of the third paragraph, noting the major difference between geo-intervention modelling and traditional Spatial Decision Support Systems (SDSS). We carefully considered the suggestion to introduce a separate literature review section and opted to retain the current structure. The references provided in the second paragraph direct readers to multiple literature reviews that already cover the main gaps in traditional SDSS, and an additional literature review may not be necessary and possibly redundant. However, we truly appreciate the suggestion and the time you have taken to review our manuscript. These changes are reflected in the tracked changes document.

Comment 3.3: In Lines 425-435, the authors state that the 80x80 grid was selected as the best model because it had the lowest MAE (82.95) compared to the 10x10 grid (MAE 1241.93). Aggregating data into larger cells naturally reduces variance and smooths the error, but it also loses the spatial granularity required for actual infrastructure interventions.

Why is a coarser scale superior for intervention (“what”) rather than just for prediction (“where”)? I suggest the authors provide more statistical and empirical evidence.

Response 3.3: Thank you for identifying this caveat in the case study. We wish to clarify that the 80x80 grid represents the finest spatial resolution evaluated, with 6,400 total cells each covering a smaller geographic area than those of the coarser 40x40 (1,600 cells) or coarsest 10x10 (100 cells) grids, as now clarified explicitly in Section 2.3.1. The lower MAE at finer resolutions reflects, in part, that smaller cells naturally contain fewer per-cell collision counts, producing smaller absolute errors independently of relative model quality. As now stated in Section 3.2, performances improved with higher cell counts from the coarser 40x40 to the finest 80x80 grid, with the greatest improvement seen from the coarsest 10x10 to the finer 40x40 grid. A meaningful cross-resolution comparison of model performance would therefore require normalizing MAE by the mean per-cell collision count at each resolution, rather than comparing absolute MAE values directly. We acknowledge that the 80x80 model was selected based on the lowest absolute MAE following the case study’s simplified demonstration design, without applying this normalization. We have expanded Appendix B.1 to note this limitation explicitly, and to clarify that the finest spatial resolution (80x80) also offers the greatest precision for locating geo-interventions. The 40x40 *tpot* model may represent an alternative that balances performance and computational cost for applications where a coarser resolution is acceptable. We also discuss MAUP in the expanded Appendix B.1, acknowledging that different resolutions may yield different variable relationships and geo-interventions, and that a systematic multi-resolution sensitivity analysis is an important consideration for practitioners applying the framework. These changes are reflected in the tracked changes document.

Comment 3.4: In Lines 456-458, the authors limit optimization to the top three most important modifiable infrastructure variables. However, is there a necessary capacity constraint? For example, can a grid cell physically accommodate 10 additional transit shelters?

I think a formal feasibility constraint should be integrated into the framework, which functions together with the min/max ranges implied in the optimization. How do you think about this?

Response 3.4: This is a valid and important concern. The framework formally accommodates physical capacity constraints through the minimum and maximum action value bounds $A_{i,j}^{min}$ and $A_{i,j}^{max}$ defined in Equation (12), which are implemented via the constraint sub-component of the geo-intervention generation component. We have added text to Section 2.2.3 in the tracked changes document making this connection explicit. In the case study, the bounds for each action variable were derived dynamically from the observed minimum and maximum values of that variable across the input dataset, meaning the optimization proposed changes only within the empirically observed range of infrastructure counts in the study area, as noted in the addition to Section 2.3.3. We agree that more explicit physical capacity

constraints per grid cell, such as the maximum number of transit shelters physically accommodable within an area based on geometric capacity or engineering guidelines, represent an important consideration beyond the demonstration scope of this case study. We have added a sentence to Appendix B.3 acknowledging this, noting that such constraints would require domain-specific knowledge and physical survey data to define appropriate per-location limits.

Comment 3.5: In Lines 463-464, the authors note that in scenario two, none of the iterations resulted in a reduction in predicted MVC. Does it mean that the model cannot apply to areas where infrastructure is currently absent? This limitation needs to be highlighted in Section 4.

Response 3.5: Thank you for raising this point. We included an additional subsection under the discussion (Section 4.3.3, Causal Inference), where scenario two is discussed as a concrete illustration of the causal inference limitation. Grid cells constrained to below-average infrastructure counts contain few training instances of those infrastructure variables co-occurring with collision outcomes, meaning the model was asked to predict the effect of infrastructure additions it has not observed during training. The optimizer encountered a prediction landscape in which no combination of changes produced a reduction in predicted MVCs across all 1101 iterations. Thus, the reliability of the generated geo-interventions is weakest where infrastructure is scarce or absent and training data for this are not available.

Comment 3.6: As for Figure 7, the geographic information reflected in Figure 7 is insufficient. The maps are too small and too simple to see the actual “what” of the intervention. For example, the authors should provide zoomed-in panels showing specific cells where the red light cameras were increased and explain if that location makes sense geographically.

Response 3.6: Thank you for this suggestion. We agree that it would be incredibly interesting to zoom in and analyze specific cells to examine and explore them in detail. However, the scope of the case study was to simply demonstrate application of the framework for understanding and focuses on modelling rather than exploratory analysis. In addition, the framework has limitations in causality that have not been accounted for, and information for examining cells in detailed (e.g. documented intervention or policies at certain cells) have not yet been collected to justify and evaluate an analysis of this level. We wish to clarify that Figure 7 was intended to provide a general sense of the spatial distribution of geo-interventions across the study area as a demonstration of the framework's output, rather than to support a detailed geographic analysis of specific cells. Also, in Section 4.3.3 (Causal Inference), the generated geo-interventions represent hypothesis candidates grounded in predictive associations rather than empirically validated recommendations. In the absence of confirmed intervention data and causal validation, interpreting the geographic plausibility of changes in specific cells would risk overstating the reliability of the framework's outputs at this stage. A detailed cell-level geographic analysis, including zoomed-in panels and assessment of location-specific plausibility, would be more appropriately situated in a focused and thorough follow-up study that incorporates the causal validation approaches outlined in Section 4.3.1. We appreciate the suggestion and recognize it as a valuable direction for future work.

Comment 3.7: Which type of model was ultimately selected in AutoML? The authors mentioned that auto-sklearn python package was used (Lines 327-332), but it is better to provide specific model information to improve the transparency of AutoML and the reproducibility of this research.

Response 3.7: Thank you for this pointing us to this missing detail. We agree that reporting the specific model selected by *autosklearn* improves the transparency and reproducibility of the case study. We have added text and a summary table (Table C1) to Appendix C identifying the selected model as a Linear Support Vector Regression (Linear SVR) from scikit-learn, along with its optimized hyperparameters as identified during the automated search. Section 3.2 has also been updated to direct readers to Appendix C for these details. These changes are reflected in the tracked changes document.

Comment 3.8: Is it possible to perform a back-testing validation? For example, train the model on 2010–2015 data, use the framework to predict the impact of infrastructure changes actually made by the City of Toronto between 2016 and 2019, and compare these predictions to the actual observed collision counts in 2019.

Response 3.8: Thank you for the excellent suggestion. We have incorporated a back-testing validation as an approach discussed the expanded Section 4.3.1 (Geo-intervention Verification). We describe this approach in general terms as training the framework on data from an earlier period and comparing the resulting geo-interventions against changes subsequently implemented and observed outcomes in a later period, following established observational before/after designs for evaluating traffic engineering measures. The Toronto case study was used as an illustrative example, where this could involve training on collision and infrastructure data from an earlier period (e.g., 2010 to 2015) and comparing the resulting geo-interventions against infrastructure changes implemented and outcomes observed in a later period (e.g., 2016 to 2020). We wish to clarify that this analysis was not conducted within the scope of the current paper, which focused on demonstrating the framework methodology, but we appreciate the suggestion and recognize it as a clear and actionable priority for future work.

Reviewer 4

Comment 4.1: This paper’s basic idea is as follows. Take various spatial datasets (e.g. accidents, roads, traffic cameras, transit shelters, traffic volumes, etc) put them into common grid cells, and train a model to predict how many accidents happen in each cell. Then, artificially change some selected variables, e.g. the number of red-light cameras or transit shelters and check whether the model predicts fewer accidents. The “geo-intervention” is basically the set of such changes that gives the best predicted improvement.

In plain terms, the authors use geospatial data and run some machine learning algorithms over it to ask: “If we changed certain city features in certain areas, would the model predict fewer accidents?”

While this sounds tempting, the main caution is that this shows what improves the model prediction, not necessarily what would truly reduce accidents in the real world. Because precise intervention data were not available, the authors could not verify that the generated geo-interventions would be feasible and effective in the real world.

Response 4.1: Thank you for taking the time and effort to review our paper. We appreciate the plain language summary and are happy to know that the idea was understood clearly. We agree that this caution is an important one and should be addressed. We included an additional subsection under the discussion (Section 4.3.3, Causal Inference) that acknowledges that the outcome modelling component learns associations from observational data rather than causal effects. Thus, the generated geo-interventions represent potential actions predicted to be *associated* with improvements rather than actions proven to *cause* those improvements. This framing is stated as having direct implications for the framework's interpretation and application, and practitioners are advised to treat the outputs as hypothesis candidates requiring causal validation. The subsection further suggests quasi-experimental and causal machine learning approaches that future extensions of the framework could adopt to move closer to estimating true real-world effects. We also note examples of extending the framework for feasibility in Section 4.4.1: "Incorporating LLMs as optional sub-components for knowledge support will also improve the lack of contextual reasoning (e.g. ranking feasibility of actions from variables, evaluating practicality of geo-interventions, suggesting approaches for outcome modelling and spatial feature engineering)". We also acknowledge the issue of feasibility in Section 4.1.3 by noting that we still require human judgement to carefully identify variables, metrics, and constraints to explore during the optimization process: "For example, non-experts can collaborate with engineers or developers to evaluate what data is used for the spatial data component, what considerations should be incorporated into the metrics of the outcome modelling component, and what actions and geo-intervention areas are likely to be effective or feasible.". All changes are available in red in the attached tracked changes document (automatically named *author-coverletter-55952670.v1.docx*).

Comment 4.2: The text is very extensive and a lot of topics have been covered. However, after all the text, and given the fact that real world data needed to evaluate the performance of the method, was not available, it all seem to be left "hanging in the air". This leaves the practical contribution difficult to assess.

Response 4.2: We appreciate this assessment and agree that the paper has not discussed practical contributions of the framework. We have addressed and acknowledged this directly by expanding Section 4.3.1 (Geo-intervention Verification) to outline several approaches through which real-world validation of the framework's outputs could be pursued, spanning retrospective back-testing, natural experiments using spatial difference-in-differences, and prospective designs from small-scale pilots to phased rollouts across spatial clusters. We hope this addition acknowledges the framework's practical limitations more clearly by demonstrating that, while real-world validation could not be performed within the present scope, the approaches to pursuing it are achievable with available or near-available data.

Comment 4.3: Since a lot of effort has been invested into producing the paper (along with all that proceeds it) I would suggest to make another additional effort and to include a special section where very specific potential ideas how to test/use the framework would be elaborated upon. It should go beyond a

general future work elaboration and describe concrete possible validation settings, datasets, or pilot studies.

In such a case I would vote the paper be accepted for publishing, but not before that.

Response 4.3: Thank you. We appreciate the time and effort you have taken to review our manuscript, and offer valuable feedback pertaining to the causal limitations of our framework. We have expanded Section 4.3.1 (Geo-intervention Verification) to include a dedicated discussion of several specific approaches through which the framework could be tested and validated, beyond a general future work statement to describe specific validation settings and general datasets using the case study as an example. These include: (1) retrospective back-testing using a temporal holdout on historical data, with the Toronto case study provided as a specific illustrative example; (2) a natural experiment approach using spatial difference-in-differences applied to independently occurring, policy-driven infrastructure deployments, cross-referenced with the framework's generated geo-interventions to identify areas of alignment; and (3) prospective designs ranging from small-scale pilot implementations to a systematic phased rollout across spatial clusters, with outcome monitoring supported by existing collision records and documented deployment data from open data portals and news sources, and collaboration with municipal authorities as an option to further strengthen the evaluation. We hope this addresses your concerns, and provides enough detail to demonstrate that the framework's outputs can be grounded in real-world causal evidence through current causal inference methods.

Reviewer 5

Comment 5.1: This paper proposes a three-component framework for generating spatially-located interventions ("geo-interventions") from observational spatial data, combining AutoML outcome modelling with Bayesian optimization to identify infrastructure changes that minimize a target variable. The framework is demonstrated through a Toronto traffic collision case study, where adjustments to red light cameras, transit shelters, and wayfinding structures are predicted to reduce motor vehicle collisions by 5.7%. The claimed novelty is reframing spatial decision support from where to intervene to what the intervention should be. Below are my comments.

Response 5.1: Thank you for your time and effort in reviewing our paper. We especially appreciate the feedback from a causal lens, as it helped us incorporate a section that addresses a very interesting and fundamental limitation of the framework. Each of the specific concerns raised has been carefully considered and addressed in the responses below, with all changes visible in red in the attached tracked changes document (automatically named *author-coverletter-55952554.v1.docx*).

Comment 5.2: The most serious problem is causal. The AutoML model learns correlations in observational data; optimizing those correlations does not predict what will happen if the infrastructure actually changes. Red-light cameras are disproportionately installed at high-collision intersections. The positive association between these cameras and MVCs in training data likely reflects placement policy rather than a safety effect. Treating permutation variable importance from a predictive model as a guide

to effective intervention is a methodological error that runs through the entire geo-intervention generation step. The authors must either incorporate causal identification methods (e.g., spatial difference-in-differences, propensity-score-weighted regression using available historical pre/post data) or explicitly bound the framework's claims to exploratory hypothesis generation rather than decision-support recommendations.

Response 5.2: We really appreciate the time and effort in identifying this important limitation of our framework and for providing detailed direction to address it. We have incorporated an additional subsection under the discussion (Section 4.3.3, Causal Inference) to explicitly bound the framework's claims to exploratory hypothesis generation rather than decision-support recommendations. The subsection acknowledges directly that optimizing associations in observational data does not estimate the effect of physically changing infrastructure, and that treating permutation importance as a guide to effective intervention is subject to the confounding flaw the reviewer identifies. The red light camera example is addressed specifically: the association with MVCs in the training data is likely attributed to placement policy installing cameras disproportionately at already high-collision intersections, a well-established form of selection bias in road safety evaluation. Transit shelters and wayfinding structures are similarly discussed as likely proxies for high-activity areas rather than direct causal levers. The subsection further includes the suggested quasi-experimental designs, specifically spatial difference-in-differences and propensity-score-weighted regression using historical pre- and post-intervention data, as potential causal identification approaches that future applications could adopt. It also discusses causal forests as a potential extension of the outcome modelling component. The known challenges of such extensions, including data availability, spatial interference, and unobserved confounding, are also mentioned.

Comment 5.3: For a GIScience paper built on spatial aggregation, the absence of any Modifiable Areal Unit Problem discussion is difficult to accept. Model performance varies dramatically across grid resolutions, yet the authors attribute this entirely to cell count and select the 80×80 grid as best without examining resolution-dependent spatial effects. The choice of grid directly shapes which infrastructure-outcome relationships the model learns and, by extension, which geo-interventions the optimizer produces. This requires analysis, not just a table of MAE values.

Response 5.3: Thank you for raising this critical point. The case study was oversimplified as a table of MAE values for understanding and demonstration, fully omitting MAUP as a common issue to consider and analyze in more thorough studies. Thus, we acknowledge the absence of a MAUP discussion as a significant gap in the original manuscript. In response, we have expanded Appendix B.1 (Spatial Feature Engineering Considerations) to include a dedicated MAUP discussion as a spatial aggregation consideration within the geo-intervention modelling framework. The addition describes the two interrelated MAUP effects: the scale effect, where results vary across levels of spatial granularity, and the zoning effect, where results vary depending on boundary configurations at the same scale, and explains how both apply to the framework's outputs. The Toronto case study was used as an illustrative example, noting the substantial variation in MAE across grid resolutions. We also imply that a systematic comparison of spatial units to learned relationships and generated geo-interventions was an important consideration for practitioners applying the framework. In addition, Appendix B.1 now notes that comparisons of absolute MAE across resolutions require caution, as finer cells naturally contain smaller

per-cell collision counts, producing smaller absolute errors independently of relative model quality, and that normalizing MAE by mean per-cell count would provide a more meaningful basis for cross-resolution comparison. These additions are reflected in the tracked changes document.

Comment 5.4: Any spatial grid over a dense urban area will produce spatially autocorrelated residuals. Standard 10-fold cross-validation does not account for this, and the reported MAE values are likely optimistic. I would expect at minimum a Moran's I test on model residuals, along with a justification for non-spatial CV or a switch to block cross-validation.

Response 5.4: Thank you for bringing this to our attention. We agree with this point. However, we note that the case study's scope was for demonstration and understanding of the framework, in which a Moran's I test and more thorough analyses are outside its scope. In response, we have expanded Appendix B.2 (Outcome Modelling and Evaluation Considerations) to discuss spatial autocorrelation as a general model evaluation consideration for practitioners applying the framework to spatially structured data. This addition described three spatial cross-validation strategies, namely block cross-validation, spatial leave-one-out cross-validation, and geographic buffering, and recommended their application when using the framework. The case study was used as an example of this limitation, noting the standard approach may have produced optimistic MAE estimates. These additions are reflected in the tracked changes document.

Comment 5.5: The complete failure of scenario 2 gets one paragraph and is left largely unexplained. This is a substantive negative result. It may indicate that the framework only functions in high-signal, high-density contexts, or that the predictive model has insufficient signal in lower-infrastructure grid cells. Either interpretation has direct implications for how practitioners should apply the framework, and neither is pursued.

Response 5.5: Thank you for taking the time to bring this to our attention. We agree that this needs to be addressed. Both interpretations the reviewer raises are discussed in the new Section 4.3.3 (Causal Inference). Scenario two was discussed as a concrete illustration of the causal inference limitation. In grid cells with below-average infrastructure counts, the training data contains few instances of the infrastructure variables co-occurring with collision outcomes, so the model has little basis for predicting the outcome of infrastructure additions. The scenario highlighted a prediction landscape in which no evaluated combination of changes produced a reduction in predicted MVCs across all 1101 iterations. This was noted as evidence of the model's unreliability when applied beyond the range of its training distribution, which demonstrated a broader issue where predictive associations in observational data do not reliably generalize to interventional contexts. We also imply that practitioners are advised to treat these geo-interventions as hypothesis-generating candidates rather than validated interventions or decision support recommendations.